



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 2B: Introduction to VHDL Processes

Module 2 Overview



- Describe combinational logic using VHDL processes.
- Compare how VHDL processes operate in hardware versus simulation.
- Construct conditionals using processes.
- Compare different styles for writing combinational logic (process versus CSA).

Processes Definition



- Definition: a **process** is a **concurrent** description that contains a collection of statements, executed in a **specific order**, with respect to a **simulator**

Processes

- Process concurrent to other processes, CSAs, and instances
 - **Required** for describing circuits with **memory**
 - **Optional** for describing **combinational** circuits
 - In Hardware: Statements within each process are executed concurrently
 - In Simulation: Statements within each process are executed in order but with “illusion” of operating concurrently

Terminology

- Statements executing in order
 - Statements within a process that are executed in a specific order (this applies only to a simulator)
 - In hardware, there is no concept of “statements executing in order”
- Sequential Circuits
 - VHDL code written to describe a design that has some sort of memory
 - Flip flops, latches, counters, FSM

Process Anatomy

```
proc_name: process (sensitivity list)
    declarations;
begin
    statements;
end process;
```

Process name (optional)

specifies when to execute the process in a **simulator**

variables valid inside the process (optional)

```
PROCESS (A, B, Ci)
BEGIN
    S <= A XOR B XOR Ci;
    Co <= (A AND B) OR (B AND Ci) OR (A AND Ci);
END PROCESS;
```

Typically RHS of statement and conditionals

- A process is contained inside of an architecture

Process – Complete Example

```
-- entity omitted for simplicity
architecture behavioral of process_example is
    -- signals, if any, go here
begin

    process (A, B)
        -- variables, if any, go here
    begin
        C <= A xor B;
    end process;

end behavioral;
```